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RESEARCH

Tween Compounds as Eco-Friendly Suppressors for Acid Copper Electroplating Applications of Microvia

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Abstract

Tween (TW) compounds, on account of their environmentally benign and non-toxic characteristics, were systematically studied as prospective eco-friendly suppressants in the process of copper plating for blind hole filling. The synergistic effects between TW and other additives were verified by means of chronopotentiometry, cyclic voltammetry, and electrochemical impedance spectroscopy. The outcomes of the electrochemical tests revealed that the competitive adsorption behavior between bis-(3-sulfopropyl) disulfide (SPS) and TW is contingent upon the current density and the intensity of convection. Metallographic section analysis demonstrated that TW exhibits outstanding microvia filling capabilities for micro blind vias with a diameter of 150 μm and a depth of 80 μm. Notably, TW-80 was able to attain a filling rate of 96.15% following electrodeposition for 60 min at a current density of 2 A/dm^2. Quantum chemical calculations suggested that the relatively smaller band gap and the clustered molecular structure of TW-80 are more conducive to its adsorption onto the electrode surface, leading to the formation of a barrier layer, which in turn restrains the deposition of copper.

Keywords Tween · Suppressor · Copper electroplating · Microvia filling

Introduction

In the realm of microelectronics manufacturing, acid copper electroplating for microvia filling has emerged as a pivotal process, ensuring reliable electrical interconnections within printed circuit boards (PCBs) and advanced packaging substrates [1–4]. The quality of via filling, characterized by void—free and uniform deposition, directly impacts the performance, reliability, and miniaturization potential of electronic devices [5–7]. To achieve this, a complex mixture of additives, including suppressors, accelerators, and levelers, is incorporated into the electroplating bath [8–11]. Among these, suppressors play a crucial role in controlling the deposition rate at different surface sites, preventing over—plating at the via openings and promoting bottom—up filling.

The additives utilized in the acidic sulfate copper plating process primarily consist of chloride ions (Cl-) [12], accelerators, suppressors and levelers. These additives interact with one another to enhance the performance of resulting formed copper plating layer [13, 14, 5, 15–17]. Previous research on suppressors for acid copper electroplating has predominantly focused on polymers and surfactants with diverse molecular structures. For instance, polyethylene glycol (PEG)-based polymers have been widely studied and utilized as suppressors due to their ability to adsorb on the copper surface and inhibit the deposition reaction [18, 19]. However, their high molecular weight and complex synthesis processes often lead to issues such as poor bath stability and difficulty in waste treatment, raising environmental concerns. Polyetheramine (PEP) has also been investigated as a potential suppressor, leveraging its unique amine-containing structure that allows for specific interactions with the copper surface [17, 20]. While PEP shows promise in adjusting the deposition rate, its relatively high cost and limited biodegradability

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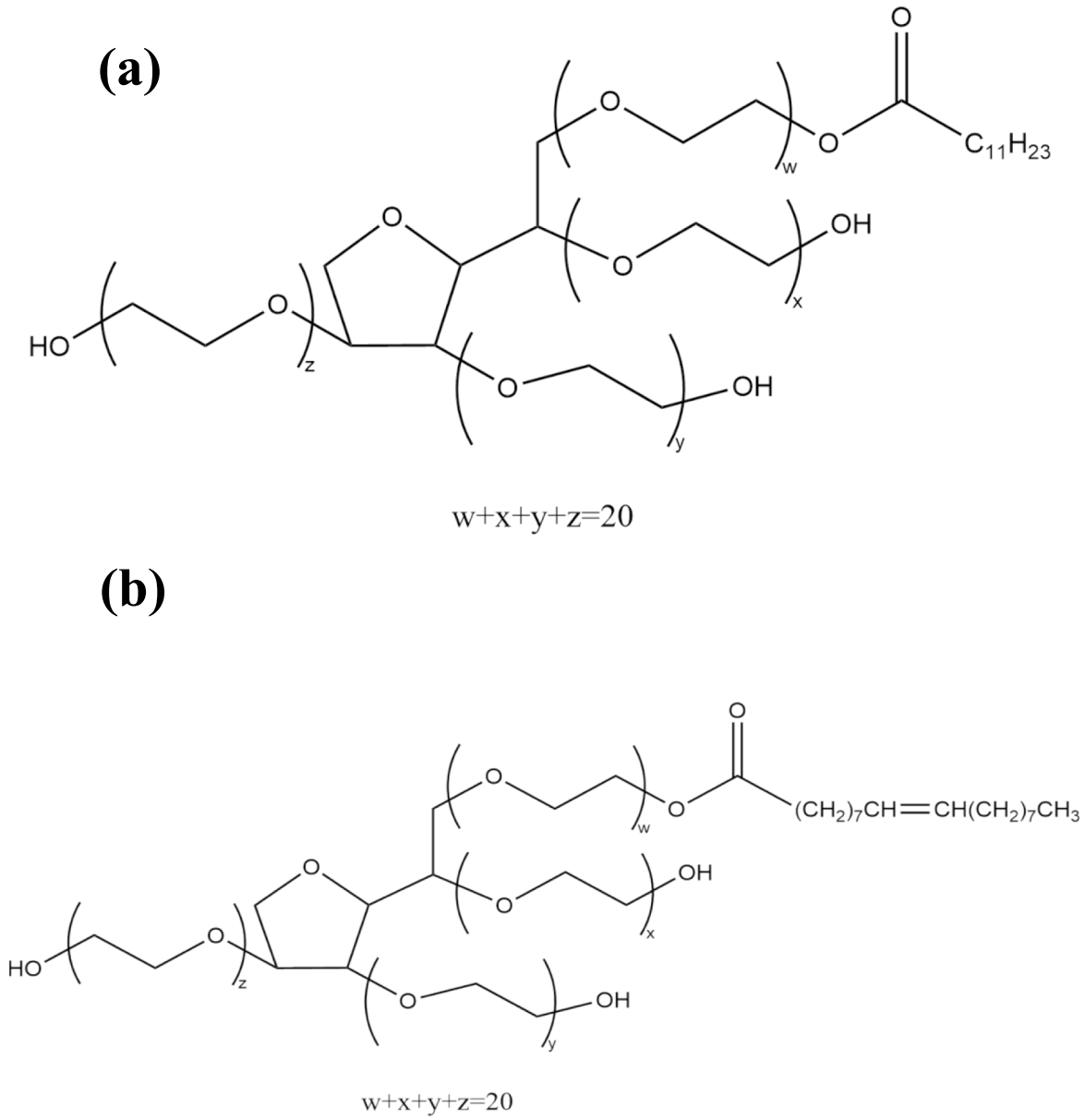


Fig. 1 Chemical structures of TW-20 (**a**) and TW-80 (**b**)

pose challenges for large—scale industrial application. Additionally, ethylene—propylene copolymer (EPE) has been explored, as its alternating structure can provide a balance between hydrophobic and hydrophilic properties, influencing the deposition behavior [21, 22]. Nevertheless, EPE may suffer from inconsistent performance under varying electroplating conditions, and its effect on the quality of microvia filling in high—aspect—ratio scenarios remains suboptimal. Moreover, some non—ionic surfactants like Triton X—series have been investigated, but they may exhibit limited efficiency in microvia filling under high—aspect—ratio conditions and can also pose potential toxicity risks.

Tween compounds, a class of non—ionic surfactants derived from sorbitan esters and ethylene oxide, have found extensive applications across multiple industries. In the pharmaceutical industry, Tween compounds are commonly used as solubilizers and emulsifiers in drug formulations. They enhance the solubility of poorly water—soluble drugs, improving drug delivery and bioavailability. For example, in the formulation of oral suspensions and injectable emulsions, Tween 80 effectively stabilizes the dispersed phases, ensuring uniform drug distribution [23]. In the food industry, Tween compounds serve as emulsifiers, preventing the separation of oil and water phases in products such as margarine, salad dressings, and ice creams. They also contribute

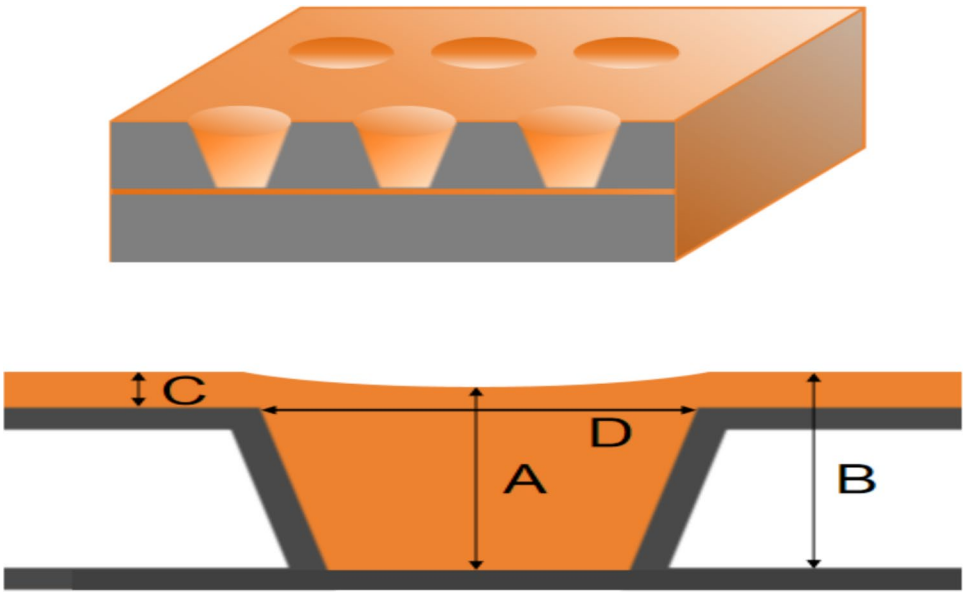


Fig. 2 A general model of microvias

to the texture and shelf—life of food products [24]. In the cosmetics industry, Tween compounds are utilized in creams, lotions, and shampoos for their excellent wetting and emulsifying properties, facilitating the dispersion of ingredients and enhancing product stability [25]. Moreover, in the field of biotechnology, Tween compounds are employed in cell culture media to reduce surface tension, promote cell adhesion, and prevent cell aggregation [26].

Against this backdrop, considering their excellent emulsifying, dispersing, and wetting properties, as well as their relatively low toxicity and good biodegradability, Tween compounds present a compelling option for electroplating applications. However, their application as suppressors in acid copper electroplating for microvia filling has remained largely unexplored. This study aims to investigate the inhibitory effects of TW-20 and TW-80 (as shown in Fig. 1) and their synergistic effects with other additives through electrochemical methods, optical microscopy, and quantum chemical calculations. By doing so, it systematically evaluates Tween compounds as eco-friendly suppressors for acid copper electroplating in microvia applications, filling this research gap. The findings of this research are expected to contribute to the development of more environmentally friendly and efficient electroplating processes, meeting the growing demands of the microelectronics industry for sustainable manufacturing.

Experimental

Electrochemical Details

The electrochemical measurements were performed with a CHI760E (ChenHua) electrochemical workstation, which was furnished with the conventional three electrodes and a 100 mL three-electrode cell capable of accommodating an virgin makeup solution (VMS) of 220 g/L CuSO4·5H2O and 28.33 mL/L H2SO4. A platinum rotating disk electrode (Pt-RDE) with a diameter of 5 mm was employed as the working electrode, while a copper strip with a diameter of 4 mm served as the counter electrode. The experiment utilized a mercurous sulfate electrode (SMSE) as the reference electrode. To evaluate the electroplating effects of the additives, the deposition potential at various rotating disk speeds was measured using chronopotentiometry. A pre-plating step was performed to deposit a 500 nm layer of copper on the platinum spinning disk electrode prior to conducting the chronopotentiometry test. The temperature of the electrolyte was maintained at a constant value of 293 ± 1 K throughout the duration of the experiment. Chronopotentiometry were conducted using different current density at 100 rpm and 1000 rpm respectively. The potential difference measured by the chronopotentiometry was defined as Eq. (1).

$$\Delta\eta = \eta_{100rpm} - \eta_{1000rpm} \tag{1}$$

Cyclic voltammetry (CV) was performed at a scan speed of 5 mV/s while maintaining a fixed Pt-RDE rotation speed of 1000 rpm. All LSV tests were conducted using a reverse scan ranging from −0.35 V to −0.80 V at a scan rate of 1 mV/s. Nyquist diagrams were generated at a constant potential of −0.65 V using an amplitude of 5 mV across a frequency range spanning from 100 kHz to 0.01 Hz [27].

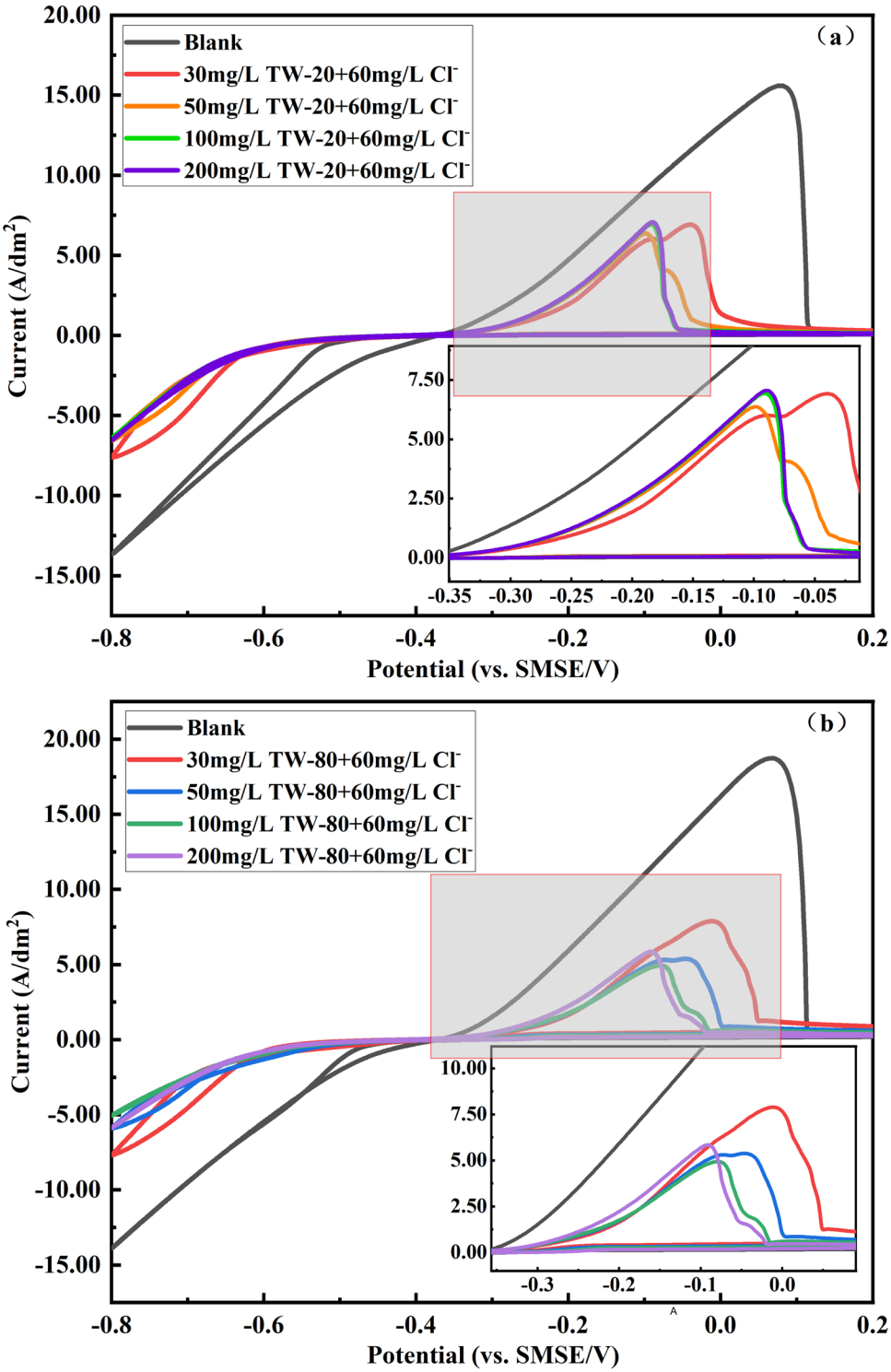


Fig. 3 CV images obtained in VMS containing quantitative Cl- by adding different concentrations of (**a**) TW-20 and (**b**) TW-80 respectively

Table 1 Integral area values (Q) of the copper dissolution peaks when both inhibitors were added stepwise to the VMS

Concentration of TW		0	30 mg/L	50 mg/L	100 mg/L	200 mg/L
Q(C)	TW-20	0.922	0.254	0.188	0.173	0.174
	TW-80	0.922	0.260	0.176	0.154	0.153

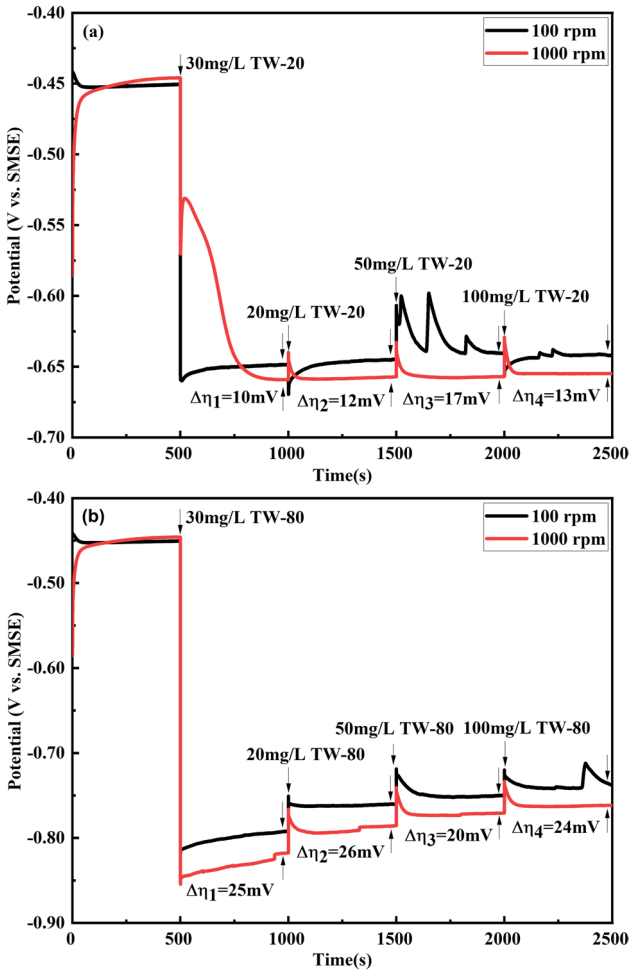


Fig. 4 Potential-time curves with the successive addition of various additives at 2 A/dm2 at two different rotation speeds: 100 and 1000 rpm

Cross Section Analysis of OM

The PCB test panels had a thickness of 1 mm, with a via diameter ranging from 120 to 150 μm and a via depth of 75 μm. Electroplating processes were conducted in a haring cell with a capacity of 1.5 L and the electroplating solution maintained a flow rate of 2.0 L per minute. The area of the cathode test plate immersed in the electroplating solution measures 50 mm × 60 mm, while the total area of the test plate is approximately 80 mm × 60 mm. In order to enhance the electrical conductivity between the copper plating and the Cu seed layer, the test boards underwent a series of decontamination processes before plating. The specific operational procedure is as follows: acid cleaning agent followed by deionized water rinse, micro-etching, another deionized water rinse, acid dipping, and finally copper electroplating in an acid bath. The electroplating test was conducted at ambient temperature with the current density maintained at 0.5, 1.0, and 2.0 A/dm2 for various plating durations. Continuous agitation was achieved by stirring the solution with flowing air bubbles to ensure thorough mixing of the plating solution and facilitate effective mass transfer in the plating cell. The composition of the plating solution included 220 g/L CuSO4·5H2O, 28.33 mL/L H2SO4, 100 mg/L TW, 6 mg/L bis(3-sulfopropyl) disulfide (SPS), 60 mg/L Cl- and 8 mg/L 1-(4-Hydroxyphenyl)−1H-tetrazole-5-thiol (HPTT). The filling rate (F) of microvias is defined by Eq. (2):

$$F = \frac{A}{B} \times 100\% \tag{2}$$

where A is the depth of the copper filled in the blind via; B is the distance from the bottom of the via to the surface of the filled copper as illustrated in Fig. 2, and C is the electroplating copper thickness on the surface of PCBs [28].

Calculation Methods

The molecular structures of TW-20 and TW-80 were optimized using the B3LYP/6-31G(d,p) method with Gaussian09 W software. The optimization of Tween's molecular structure was conducted based on frontier molecular orbital theory, focusing on the lowest unoccupied molecular orbital (LUMO) and the highest occupied molecular orbital (HOMO) [29]. The forcite module and VMD software were respectively adopted to investigate the interaction between copper surface and Tween and visualize the optimized geometry and molecular orbitals [30]. Tween molecular dynamics properties in the aqueous phase such as radius of gyration, radial distribution function and diffusion coefficient were investigated by the Material Studio software at the temperature of 298 K. The COMPASSII force field was utilized to optimize the size and simulate the adsorption process of Tween.

Results and Discussion

Electrochemical Evaluation

The study investigated the inhibitory effects of TW and identified the optimal concentration that exhibited the highest inhibitory strength, as depicted in Fig. 3. It can be observed that when there is no Cl- and TW, the anode peak current density is the highest, reaching 15.69 A/dm^{2}and 18.82 A/dm^{2}respectively. When Cl- and TW are added, the lowest values of the anode peak current density are 6.35 A/dm^{2}and 4.95 A/dm^{2}respectively. Meanwhile, the polarization potential of copper deposition obviously shifts in the negative direction.

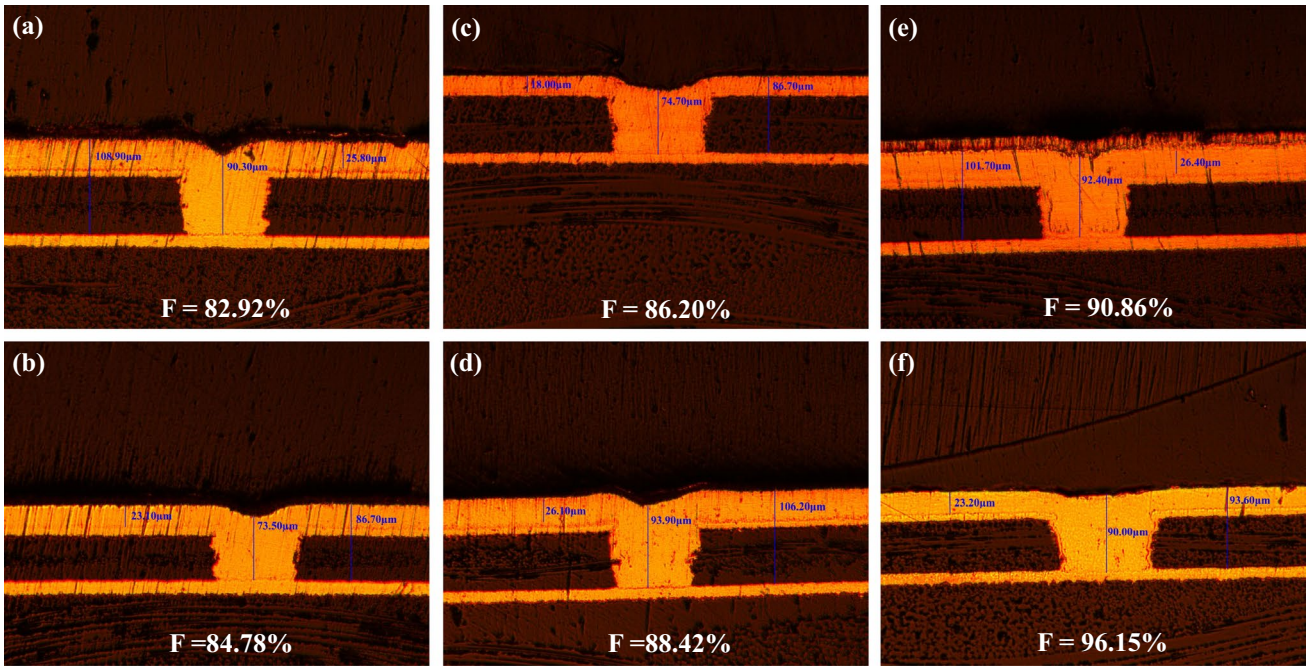


Fig. 5 Cross-section images of microvias with diameters of: **a** 130 μm for 240 min with TW-20 at 0.5 A/dm2, (**b**) 120 μm for 120 min with TW-20 at 1 A/dm2, (**c**) 150 μm for 60 min with TW-20 at 2 A/dm2, the plating solution is composed of VMS, 40 mg/L Cl-, 100 mg/L TW-20, 6 mg/L SPS and 8 mg/L HPTT; **d** 130 μm for 240 min with TW-80 at 0.5 A/dm2, (**e**) 130 μm for 120 min with TW-80 at 1 A/dm2, (**f**) 150 μm for 60 min with TW-80 at 2 A/dm2, the plating solution is composed of VMS, 30 mg/L Cl-, 100 mg/L TW-80, 6 mg/L SPS and 8 mg/L HPTT

Under the condition of a current density of 2 A/dm^2, the corresponding deposition potentials are −0.666 V and −0.678 V respectively. During the forward scanning process, the dissolution peak of copper decreases significantly. Table 1 lists the values of the integral area Q of the copper dissolution peak on the CV curve measured in the electroplating solutions containing inhibitors with different concentrations.

In general, a lower Q value signifies a more potent inhibitory effect of the inhibitor. These outcomes suggest that as the concentration of inhibitors increase, both TW-20 and TW-80 reach their maximum inhibitory strength at a concentration of 100–200 mg/L. Additionally, within the concentration range of 50–200 mg/L, TW-80 demonstrates a superior inhibitory effect compared to TW-20. The findings of this experiment align with previous research [31], which demonstrates that inhibitors with larger molecular weights exhibit stronger inhibitory capabilities when their molecular structures are consistent.

Following the determination of the inhibitory strengths of the two inhibitors, this study utilized the chronopotentiometry to predict the hole-filling capacity of the plating solution consisting of 60 mg/L Cl-, 6 mg/L accelerator, 4 mg/L HPTT and separate batches of 200 mg/L TW-20 or TW-80. The test results are shown in Fig. 4. The analysis of the test results indicated that the plating solutions containing distinct inhibitors all adhered to the convection-dependent adsorption (CDA) of additives [32], as evidenced by a notable negative shift in the stable potential measured at high rotation speeds. In other words, the inhibitors exhibit a stronger inhibitory effect under conditions of strong convection, thereby complicating the process of copper deposition. Furthermore, it can be inferred from Fig. 4 that the overpotential for copper deposition with TW-80 is higher than that with TW-20, which remains unaffected by the intensity of convection. After comparing the potential difference Δη calculated for the two inhibitors at different rotation speeds, it becomes apparent that Δη for TW-80 consistently exceeds that for TW-20 at equivalent concentrations. According to the CDA mechanism, a higher Δη of the plating solution is associated with increased hole-filling capability [32], suggesting that TW-80 is expected to have a more robust capacity for filling holes.

Combining the CV and potential-time images, the optimal concentration for both TW-20 and TW-80 was determined to be 100 mg/L. Using the same method, the optimal electroplating formulations corresponding to TW-20 and TW-80 were obtained: (a). 40 mg/L Cl-, 100 mg/L TW-20, 6 mg/L SPS and 8 mg/L HPTT; (b)0.30 mg/L Cl-, 100 mg/L TW-80, 6 mg/L SPS and 8 mg/L HPTT.

Electroplating Experiments

To assess the effectiveness of TWs in copper plating, a 1.5L plating solution was prepared and then transferred

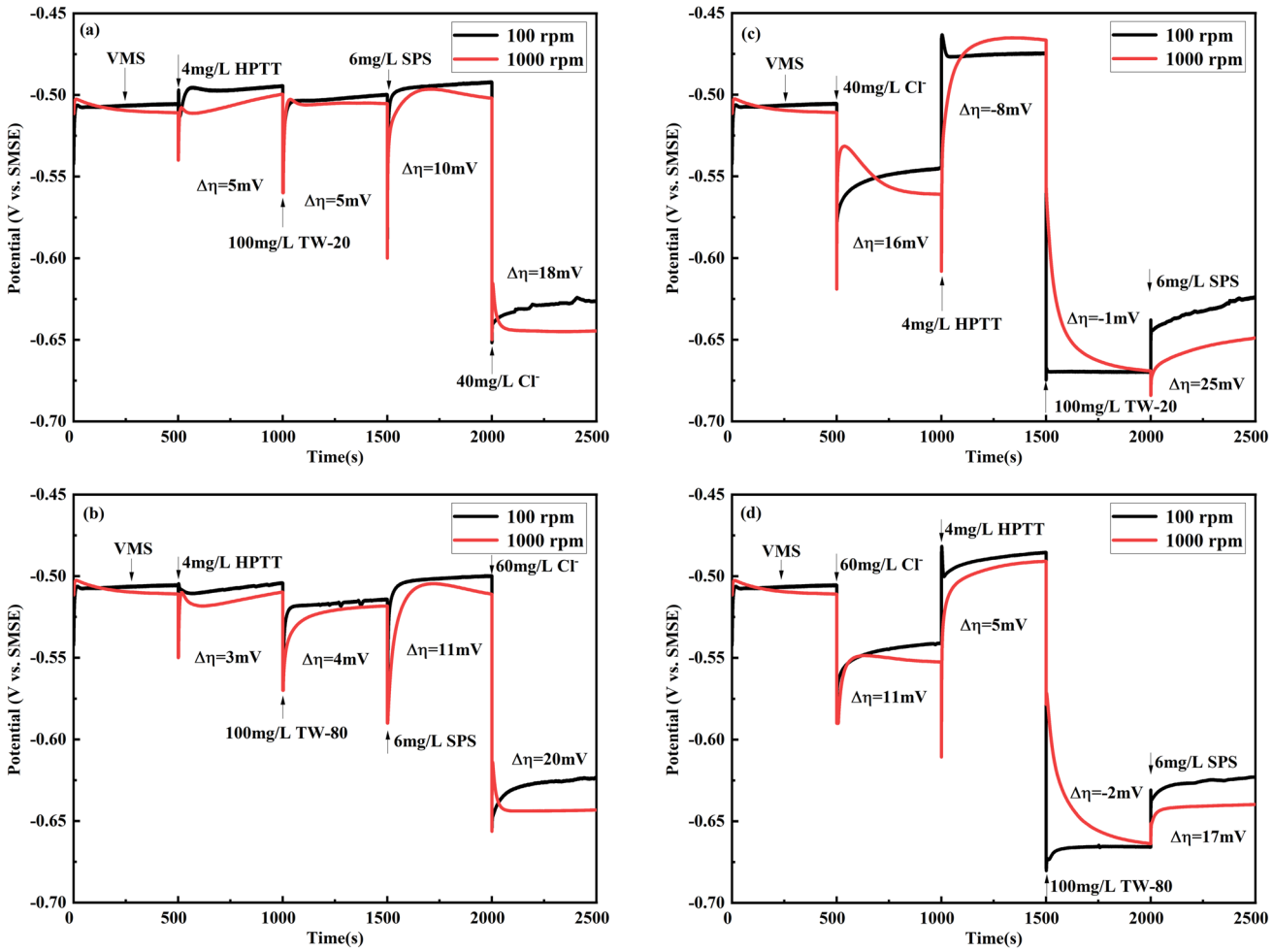


Fig. 6 Potential-time curves with the successive addition of various additives at different rotation speeds

into the plating tank. Subsequently, a test board containing blind holes with diameters ranging from 120–150 μm was immersed in the plating solution. The plating experiments were carried out using deposition current densities of 0.5 A/dm2, 1 A/dm2, and 2 A/dm2. After the metallographic sample was filled with resin, it was polished and subsequently examined and analyzed using a metallographic microscope. Cross-sectional images were obtained to demonstrate the filled blind holes under different inhibitors. Figure 5(a-f) show that an increase in current density led to a gradual decrease in the surface copper thickness for both sets of metallographic samples, while the filling rate gradually improved. This phenomenon can be attributed to the higher likelihood of the accelerator replacing the inhibitor at lower current densities, which prolongs the time required for each additive to reach adsorption equilibrium. Consequently, there is minimal difference in the rates of copper deposition between the interior and exterior of the holes over a prolonged period, which leads to a decrease in the filling rate and an increase in the thickness of the surface copper. When comparing the metallographic samples with two inhibitors at the same current density, it was consistently observed that TW-80 demonstrated superior filling performance, aligning with the results anticipated from chronoamperometry.

Electrochemical Behavior of the Additives

To further clarify the interaction of these additives, the synergistic effects of TW with other additives were further investigated through chronopotentiometry, as shown in Fig. 6(a-f). The analysis will be based on TW-80 since the structural similarity between TW-20 and TW-80. As illustrated in Fig. 6(b), the leveling agent HPTT does not influence copper deposition under either strong or weak convection intensities. Following the addition of TW, a slight negative shift in the deposition potential is observed. The subsequent introduction of SPS leads to a positive shift in the deposition potential. Finally, the addition of Cl- results in a rapid negative shift in the copper deposition overpotential

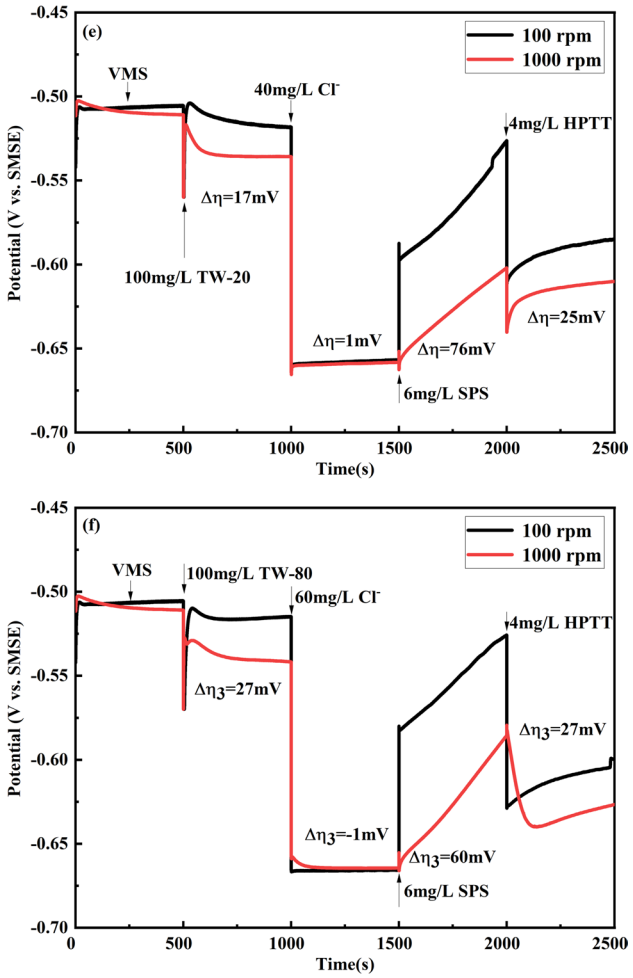


Fig. 6 (continued)

and an increase in Δη to 20 mV, indicating the occurrence of convection-dependent adsorption (CDA) behavior.

Comparing the time intervals of 500–1000 s in Fig. 6(b) with the 1000–1500 s interval in Fig. 6(d) reveals that the simultaneous presence of HPTT and Cl- can accelerate the reduction of copper ions, whereas no such effect is observed when HPTT is present alone. Furthermore, a comparison between the intervals of 500–1500 s in Fig. 6(b) and 500–1000 s in Fig. 6(f) indicates that when only HPTT and TW are present in the VMS, HPTT diminishes the inhibitory effect of TW, suggesting an antagonistic interaction between the two. This antagonism reduces the adsorption capacity of both HPTT and TW in regions of strong convection, while exhibiting no significant impact under weak convection conditions. This implies that HPTT accelerates the rate of copper deposition outside blind holes, leading to an increase in the thickness of the deposited copper.

Moreover, when analyzing the time intervals of 500–2000 s in Fig. 6(d) alongside those of 500–1500 s in Fig. 6(f), it becomes evident that the addition of Cl- disrupts the antagonistic interaction between TW and HPTT. TW and Cl- preferentially combine to form a TW-Cl- complex, significantly enhancing the inhibitory effect on copper deposition. This suggests that in the simultaneous presence of Cl-, HPTT, and TW, Cl- preferentially associates with TW, thereby augmenting its inhibitory effect.

Following the determination of the optimal concentrations of TW-20 and TW-80, they were subsequently fixed at 100 mg/L. The relationship between their inhibitory strength and Cl- concentration was then investigated using CV to ascertain the corresponding optimal Cl- concentration, as illustrated in Fig. 7(a-b). As the concentration of Cl- in the electrolyte rises, there is an initial increase followed by a decrease in the cathodic polarization of copper ion deposition, which aligns with the pattern of initially decreasing and then increasing copper dissolution peaks. Moreover, with the gradual increase in Cl- concentration, the current–potential"hysteresis loop"resulting from distinct scan directions is notably attenuated [33]. Figure 7(a) illustrated that at a Cl- concentration of 40 mg/L, TW-20 demonstrates the most pronounced inhibitory impact on copper ion deposition, whereas at a Cl- concentration of 30 mg/L, TW-80 displays the most potent inhibitory effect on copper ion deposition. Continuing to elevate the Cl- concentration not only fails to enhance the inhibitory effect of the surfactants but also accelerates copper deposition, a phenomenon associated with the depolarizing effect of Cl-. The conclusions drawn from the figure are consistent with existing literature [34], suggesting that there is an optimal Cl- concentration for a fixed inhibitor concentration that maximizes its inhibitory effect.

The inhibitory potency of TW at the optimal concentration of Cl- was examined through linear sweep voltammetry (LSV), as shown in Fig. 8. The figure clearly demonstrates that the cathodic polarization of copper ion deposition exhibits a gradual increase upon the addition of 100 mg/L of TW to the VMS solution. Nevertheless, the extent of this augmentation in polarization is currently minimal. The observation suggests that elevating the concentration of TW in the plating solution may partially augment its adsorption on the cathode surface, consequently enhancing its inhibitory effect on copper deposition. At this stage, the adsorption of TW on the cathode surface is primarily attributed to electrostatic and van der Waals forces [35], leading to a comparatively mild inhibitory effect. When Cl- was present in the solution, there was a significant increase in the cathodic polarization of copper ion deposition. This is evidenced by a depolarization effect on the copper ion deposition process. The only plausible explanation for this experimental phenomenon is the synergistic interaction between Cl- and TW. The existence of TW can enhance the adsorption of TW on the cathode surface, thereby increasing its inhibitory effect on copper ion deposition. The heightened inhibitory effect

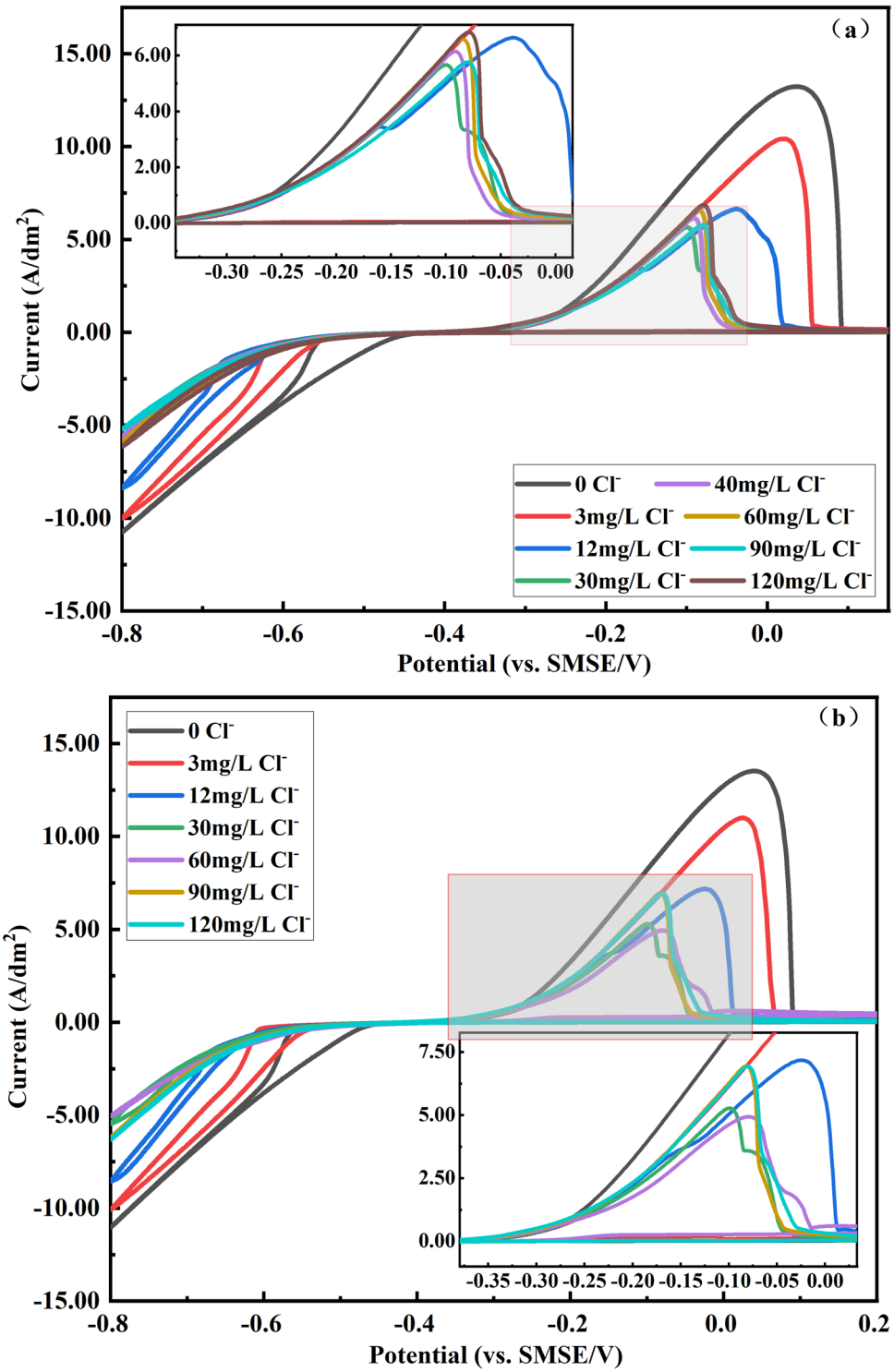


Fig. 7 CV images obtained in VMS containing 100 mg/L suppressor (**a**: TW-20, **b**: TW-80) by adding different concentrations of Cl- respectively

on copper ion deposition, as a result of the adsorption of TW molecules on the cathode surface due to an increase in Cl- concentration, is crucial to highlight. This effect is achieved through chemical bonding, ultimately leading to a notable enhancement in the inhibitory effect.

Moreover, at a current density of 2 A/dm2, the inhibitory potency of the two inhibitors is ordered as TW-80 > TW-20, which is consistent with the previously established conclusion. The starting potential, at which the current begins to rise, varies with different inhibitors as the electrode potential becomes increasingly negative. Additionally, the rate of current increase also shows variation. Based on these analyses, it can be concluded that the difference in optimal Cl- concentration between TW-20

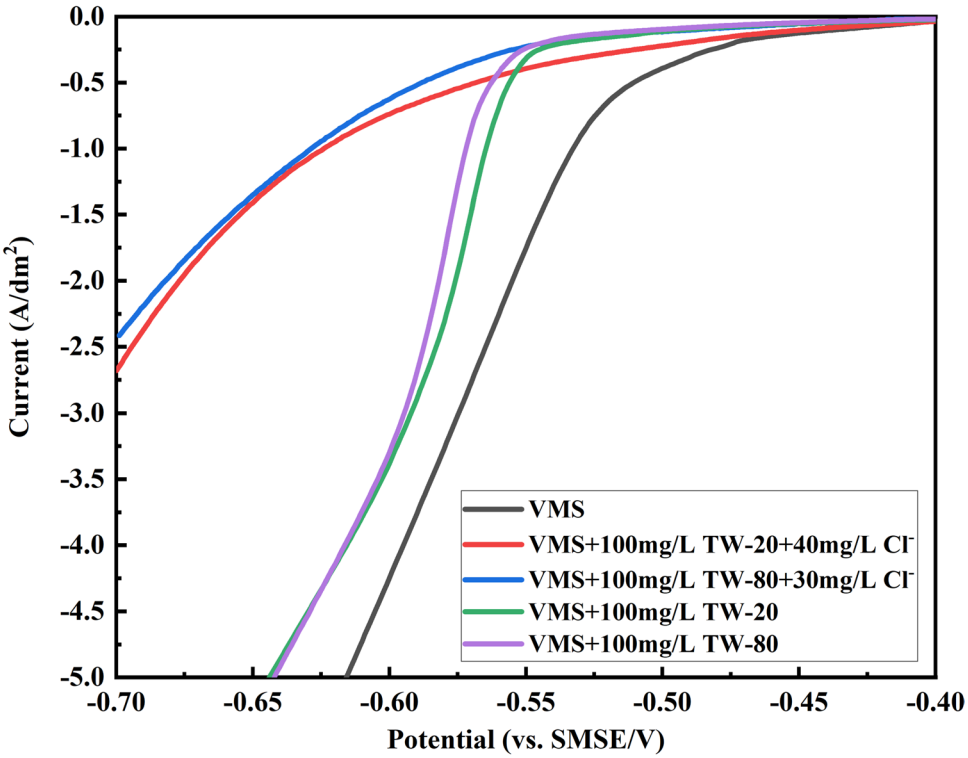


Fig. 8 Linear sweep voltammetry curve measured at 1000 rpm

and TW-80 is not statistically significant, resulting in the polarization curve trends being highly similar.

The Nyquist plots derived from the plating solutions containing TW-20, TW-80 and different concentrations of Cl- are depicted in Fig. 9, along with their corresponding equivalent circuit diagrams. It is important to note that, in order to provide a more objective reflection of the adsorption of inhibitors in the actual copper plating experiment for blind hole filling, the electrochemical impedance data were measured at a polarization potential of −0.65 V (vs. SMSE). The equivalent circuit diagram shows the different components representing resistance and capacitance in a system. Rs represents solution resistance, while CPE1 and R1 represent the capacitive arc in the high-frequency region. CPE1 represents double-layer capacitance, while R1 signifies electrochemical reaction resistance. Similarly, CPE2 and R2 correspond to another capacitive arc, indicating Cu + asd relaxation on the electrode surface [36–38]. The magnitude of R1 can indicate the challenges associated with copper ion deposition, enabling the assessment of Cl- concentration on TW's inhibitory capacity against copper deposition.

Table 2 revealed that when the VMS solution contained 100 ppm of TW-20 and 100 ppm of TW-80, an increase in Cl- concentration to 30 mg/L led to the values of *R1* increasing from 5.73 Ω cm2 and 5.51 Ω cm2 to 17.17 Ω cm2 and 16.58 Ω cm2, respectively. This finding suggests that, under specific conditions, an elevation in Cl- concentration can significantly enhance the inhibitory effect of TW-20 and TW-80 on copper deposition. Furthermore, when *R1* attains its maximum value for TW-20 and TW-80, the associated Cl- concentration aligns with the concentration recorded in prior experiments. In contrast, the absence of additives in the VMS solution results in a *R1* value of 4.004 Ω cm2. The comparison of the rise in *R1* resulting from the introduction of TW and Cl- concentrations reveals that an elevation in Cl- concentration can more efficiently amplify the inhibitory effect of TW. This phenomenon is closely associated with the adsorption mechanism of TW on the electrode surface. As previously stated, when plating solutions contain higher concentrations of Cl-, TW interacts with Cl- and subsequently adheres to the cathode surface through chemical bonding. In this instance, TW exhibits high adsorption strength, leading to a potent inhibitory impact on copper deposition. Conversely, the absence of Cl- in the plating solution results in the adsorption of TW molecules on the electrode surface through electrostatic or van der Waals forces, which leads to a less robust adsorption and a diminished inhibitory effect on copper ion deposition.

Quantum Chemical Calculations and Molecular Dynamics Simulation

In order to further explore the molecular structure and properties of TW-20 and TW-80 for optimizing the

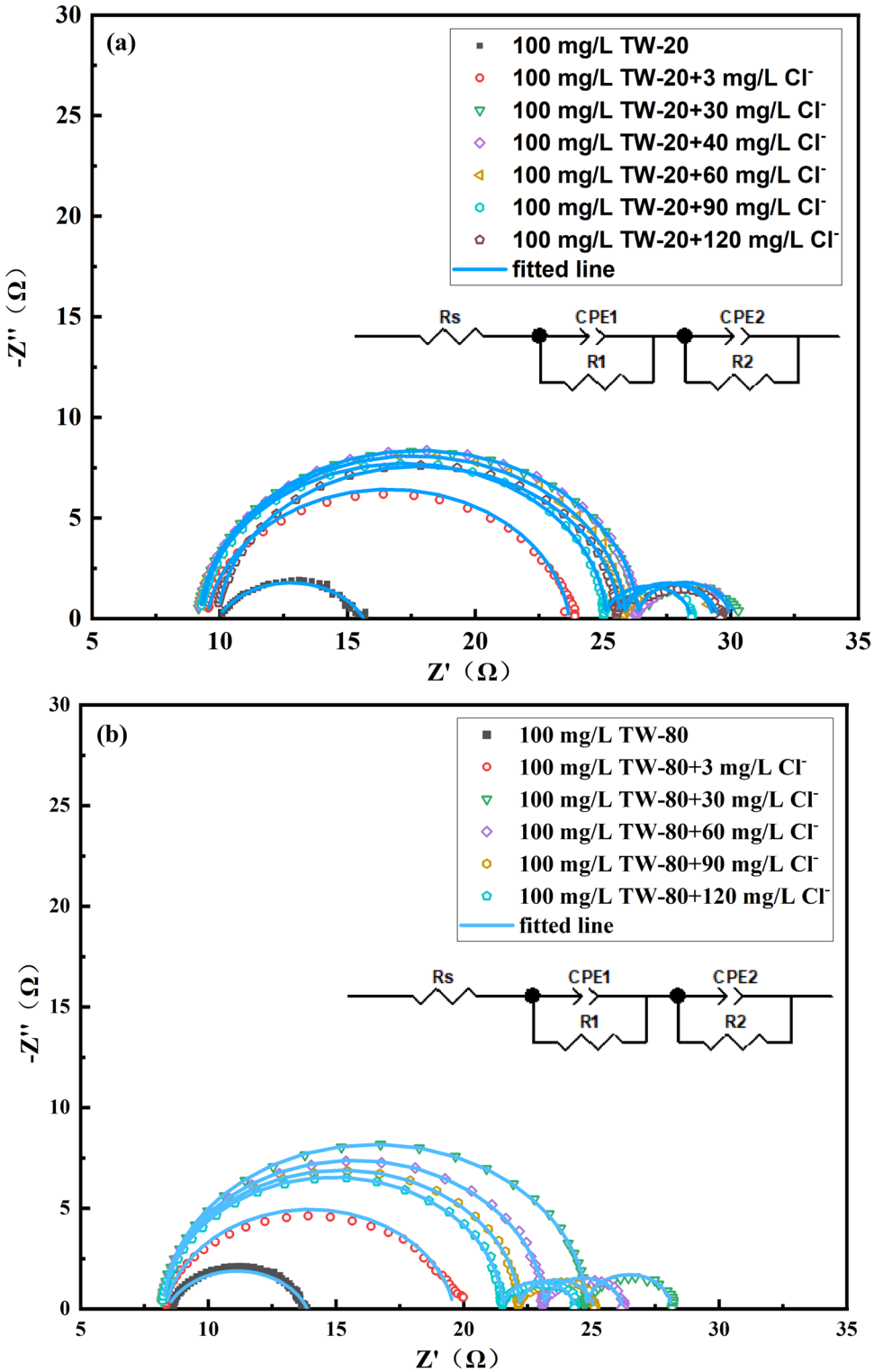


Fig. 9 Nyquist was carried out at 1000 rpm and −0.65 V with electrolyte composed of 0.88 M CuSO4 5H2O, 0.54 M H2SO4, 100 mg/L (**a**) TW-20/ (**b**) TW-80 and Cl- in different concentrations

adsorption model, quantum chemical calculation methods were employed in this study.

The adsorption behavior of TW-20 and TW-80 molecules on the Cu(111) surface was simulated using the Dynamic module of the Materials Studio software, and the results are illustrated in the Supplementary Material Fig. 1. It can be observed that TW-20 exhibits a linear structure on the copper surface, while TW-80 forms a clustered structure. Compared to the linear configuration of TW-20, the clustered arrangement of TW-80 promotes more favorable adsorption

Table 2 Electrochemical impedance parameters in TW-20 and TW-80 electroplating solutions at various chloride ion concentrations

	Concentrations of Cl- (mg/L)	Rs (Ω cm2)	R1 (Ω cm2)	R2 (Ω cm2)
TW-20	0	9.907	5.73	
	3	9.495	14.19	
	30	9.217	17.17	3.821
	40	9.226	17.16	3.579
	60	9.293	16.59	3.396
	90	9.304	15.82	3.320
	120	9.945	15.70	3.872
	0	8.364	5.51	
TW-80	3	8.318	11.3	
	30	8.242	16.58	3.302
	60	8.242	14.92	3.039
	90	8.249	13.93	2.906
	120	8.252	13.27	2.848

on the electrode surface, resulting in the formation of a dense blocking layer with a thickness of approximately one nanometer.

Models comprising 500 water molecules and a copper atomic layer were assembled utilizing Material Studio software. The Forcite Energy module was then employed to compute the adsorption energy of TW-20 and TW-80 molecules on the copper layer in a solution without Cl-, as shown in Table 3. The findings suggest that the adsorption energy of TW-80 molecules is considerably more negative than that of TW-20 molecules. This implies that in the absence of Cl-, the binding energy between TW-80 molecules and the copper layer is significantly higher than that of TW-20 molecules, which promotes the adsorption of TW-80 molecules onto the copper layer.

The charges of atoms in TW-20 and TW-80 molecules were calculated using Gaussian09 W and GaussView5.0 software. Furthermore, frontier molecular orbital diagrams and ESP images were generated. Due to limitations in the software's ability to accurately calculate charges for large

Table 3 Adsorption energy of TW-20 and TW-80 molecules on copper layers in aqueous solution

TW-20	Energy(eV)
E(Cu + TW-20)	1665.085
E(TW-20)	7.355
E(Cu)	1657.924
Eads	−0.194
TW-80	
E(Cu + TW-80)	1657.588
E(TW-80)	6.519
E(Cu)	1657.924
Eads	−6.855

Table 4 Adsorption energy of TW-20 and TW-80 molecules on copper layers in aqueous solution

Suppressor molecule	EHOMO (eV)	ELUMO (eV)	ΔE (eV)
TW-20	−6.642	0.160	6.802
TW-80	−6.330	0.161	6.491

molecular compounds, the molecular structures of TW were simplified for the purposes of this experiment. The findings are presented in supplementary material as Fig. 2 (a)-(f), Fig. 3 and Table 4 below. According to the theory of molecular frontier orbitals, the HOMO (highest occupied molecular orbital) and LUMO (lowest unoccupied molecular orbital) play a crucial role in the adsorption of organic molecules on metal surfaces [39]. A smaller ELUMO value signifies a greater electron-accepting capability, whereas a larger EHOMO value indicates a stronger electron-donating capacity [40, 41]. A narrower band gap (ΔE = ELUMO-EHOMO) indicates higher molecular activity, which is advantageous for adsorption. The data from Table 4 indicates that the ΔE value for TW-80 is comparatively smaller than that for TW-20, suggesting a higher propensity for TW-80 molecules to adsorb onto the copper surface. The ESP image offers a direct visualization of favored reaction sites, with the red region indicating low electron density and the blue region indicating high electron density. In general, atoms located in regions of high electron density exhibit a stronger tendency to form bonds with metal surfaces [42, 43]. Figure 2 in the supplementary materials indicates that TW-20 and TW-80 molecules possess a comparable distribution of electron density regions as a result of their structural similarity, with both demonstrating elevated electron density at the oxygen atoms. The sole distinction lies in the higher electron density at the carbon–carbon double bond in TW-80 molecules. The higher electron density suggests that the double bond carbon atoms in TW-80 molecules are more prone to forming a carbocation with hydrogen ions in the acidic plating solution. Subsequently, they are more likely to undergo electrophilic addition with Cl^-, resulting in chloride adsorbed the copper surface.

Based on the aforementioned research findings and pertinent literature [20, 23], this study puts forward an adsorption model for TW-20 and TW-80, as shown in Fig. 13. In the absence of Cl- in the plating solution, TW molecules predominantly adhere to the electrode surface via electrostatic interactions and van der Waals forces, as illustrated in Fig. 3(a1) and (b1) in the supplementary materials. In this scenario, the intermolecular forces between TW molecules and the electrode surface are relatively weak, resulting in a reduced inhibitory effect of the resulting barrier layer on copper deposition. Upon the addition of Cl- to the plating solution, TW molecules engage with Cu + and Cl- to create complexes (TW-Cu

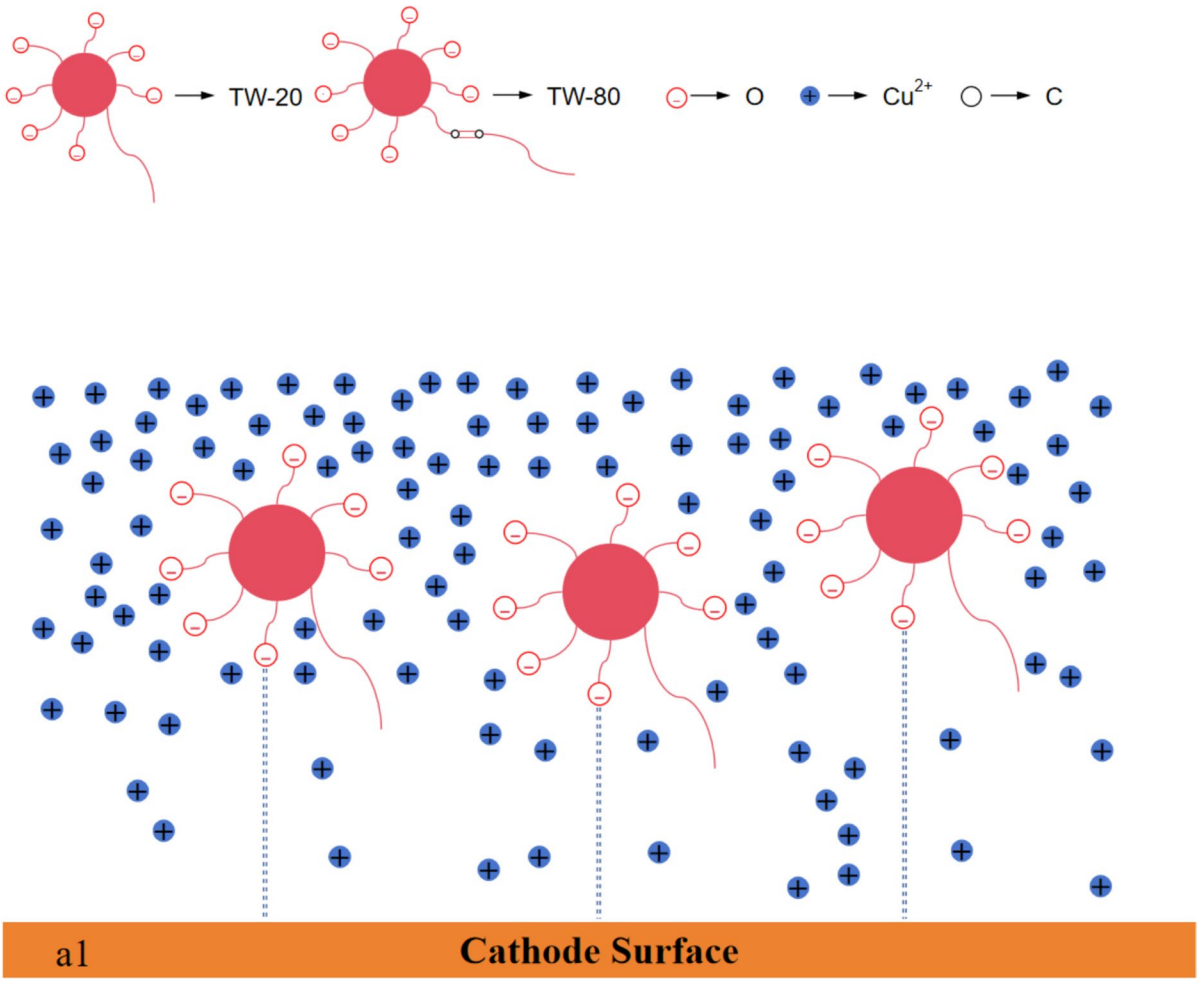


Fig. 10 The optimized adsorption model of TW-20 (**a**) and TW-80 (**b**) on the cathode surface

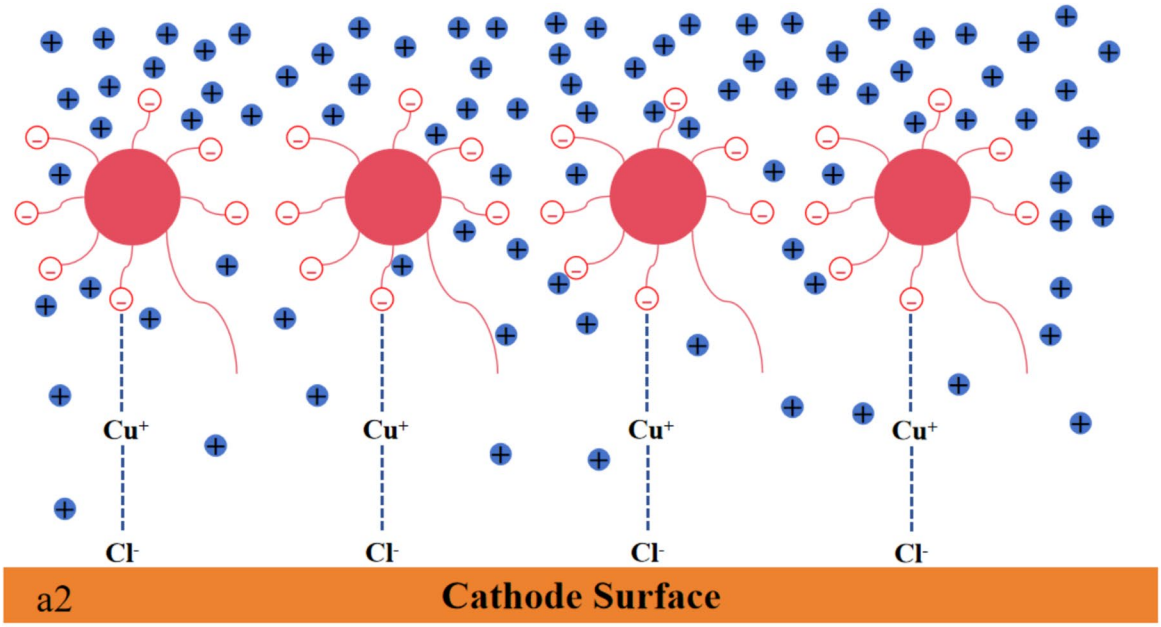
+-Cl-). Consequently, it is not just simple TW molecules, but rather these complexes that adsorb on the electrode surface. The TW-Cu + -Cl- complexes adhere to the electrode surface via chemical bonding, resulting in a more potent inhibitory impact on copper deposition due to the formation of a barrier layer. The carbon–carbon double bond in TW-80 molecules specifically undergoes a reaction with Cl- and subsequently adsorbs onto the electrode surface through Cl-. This process leads to the formation of a barrier layer by the TW-80-Cu + -Cl- complex, which exhibits a more potent inhibitory effect in comparison to the barrier layer formed by the TW-20-Cu + -Cl- complex (Fig. 10).

Conclusions

The study shows that TW-20 and TW-80 have inhibitory effects on copper deposition, making them suitable for use as electroplating inhibitors. TW-80 demonstrated exceptional performance, with a plating solution comprising 220 g/L CuSO4·5H2O, 28.33 mL/L H2SO4, 30 mg/L Cl-, 6 mg/L SPS, 8 mg/L L, and 100 mg/L TW-80 achieving full filling of blind holes with a 150 μm diameter, resulting in a filling rate of 95.6%.

Electrochemical measurements have uncovered a convection-dependent synergistic effect between TW and L at low current densities. Moreover, a significant synergistic inhibitory effect was observed between TW and Cl-, suggesting that Cl- amplifies the inhibitory effect of TW on copper deposition. The coexistence of TW, L, and Cl- led to the most potent inhibitory effect.

Quantum chemical calculations revealed that TW-20 molecules adopt a linear configuration in aqueous solution, whereas TW-80 molecules tend to form a cluster structure, rendering them more susceptible to adsorption on the electrode surface. This results in the formation of a dense and relatively thick barrier layer. The binding energy between TW-80 molecules and the copper layer exhibited a significantly higher magnitude compared to that of TW-20



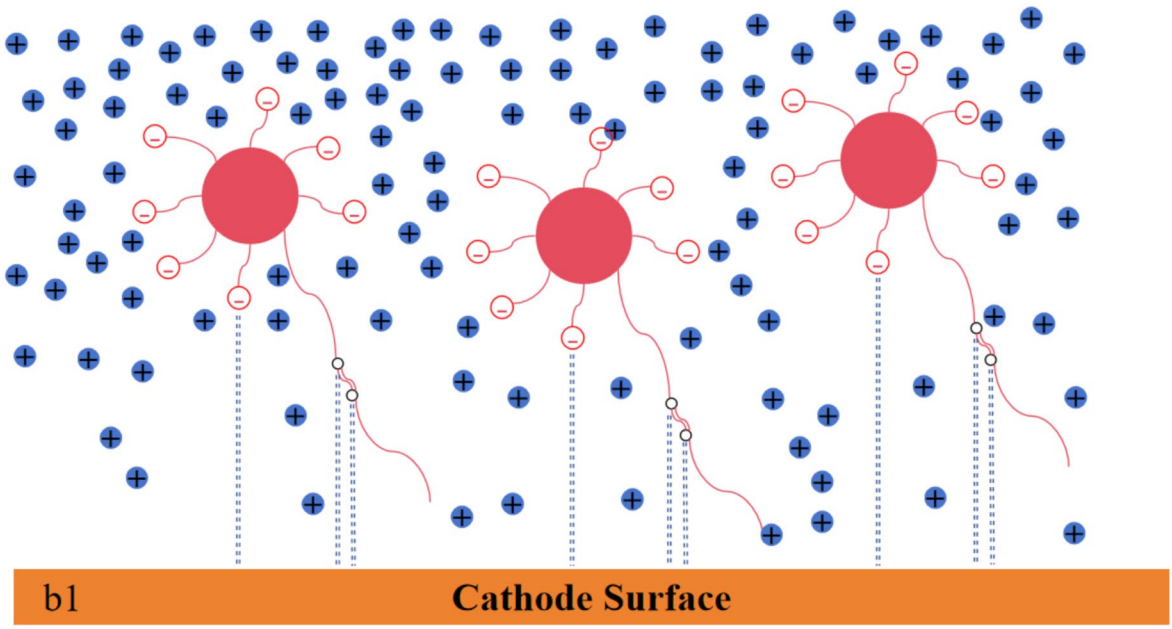


Fig. 10 (continued)

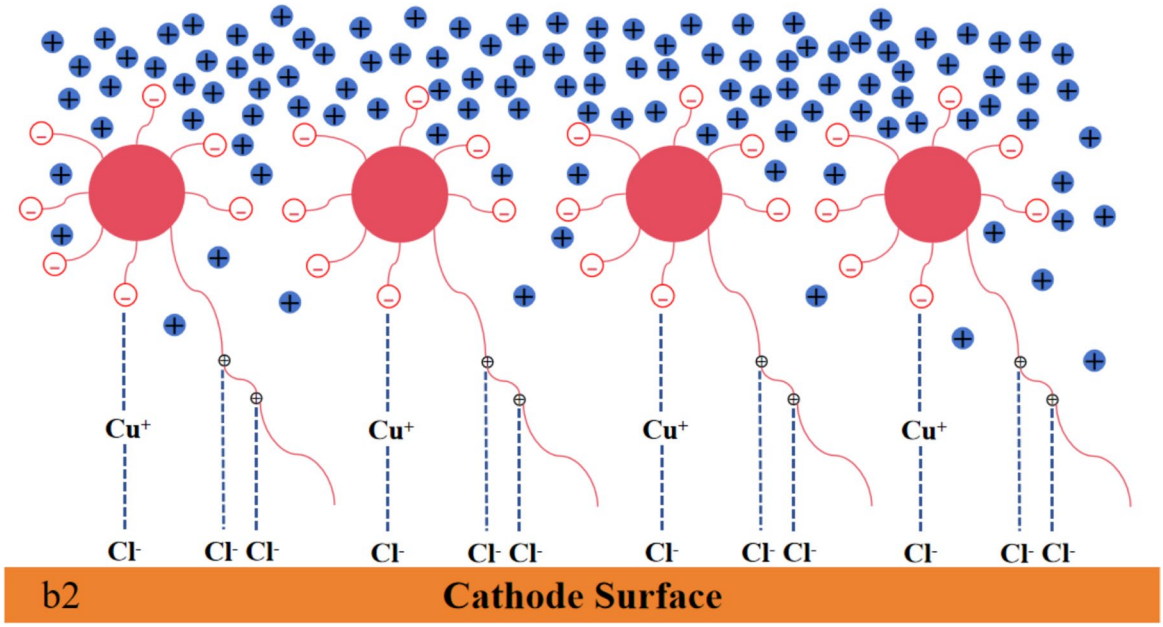


Fig. 10 (continued)

molecules, indicating a greater propensity for adsorption onto the copper layer. Moreover, the presence of a carbon–carbon double bond in TW-80 molecules was found to enhance their adsorption on the copper layer. Subsequently, the adsorption models of TW-20 and TW-80 were developed based on the aforementioned experimental results.

Based on the research on Tween compounds as eco—friendly suppressors for acid copper electroplating in microvia applications, several key guidelines are proposed. Firstly, when selecting Tween compounds, factors like their hydrophilic—lipophilic balance, molecular weight, and branching structure should be carefully considered to ensure proper adsorption and inhibitory effects. Secondly, concentration optimization is crucial, as both too low and too high concentrations can negatively impact microvia filling quality, thus requiring systematic studies to find the optimal balance. Thirdly, using Tween compounds in combination with other electroplating additives and understanding their synergistic effects can enhance the overall plating performance, although their compatibility and interaction mechanisms need thorough investigation. Finally, environmental considerations are essential, including minimizing compound loss during bath operation and adopting appropriate biodegradation processes for waste treatment. By adhering to these guidelines, Tween compounds can be effectively applied, promoting more sustainable and efficient microelectronics manufacturing, and future research should further explore and optimize their application.

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Author Contribution L.Z wrote the main manuscript text. T.Z provided ideas. H.Y prepared Figs. 1–2. All authors reviewed the manuscript.

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Ethical Approval Not applicable.

Competing interests The authors declare no competing interests.

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